

Capítulo III - Requirements Specification

En esta sección, se presenta la especificación de requisitos de AgroSafe a partir de los hallazgos obtenidos en las etapas previas de análisis del problema, entrevistas, needfinding, Big Picture EventStorming y definición del Ubiquitous Language. Su propósito es traducir el entendimiento del dominio en un conjunto estructurado de requisitos que orienten de manera clara el desarrollo del producto dentro del alcance del proyecto.

A partir de ello, este capítulo organiza los requerimientos funcionales y técnicos de la solución en artefactos que permiten vincular necesidades de los usuarios, objetivos del producto y decisiones de implementación a nivel de MVP académico. En particular, se detallan las **User Stories**, el **Impact Mapping** y el **Product Backlog**, con el fin de asegurar trazabilidad entre el problema identificado, el valor esperado para los segmentos objetivo y el trabajo que deberá ejecutar el equipo durante el desarrollo del sistema.

3.1. User Stories

En esta sección se presentan los requisitos funcionales y técnicos del sistema AgroSafe IoT mediante la definición de Epics, User Stories y Technical Stories. Las historias han sido elaboradas considerando los distintos roles que interactúan con el ecosistema (visitantes, agricultores, agrónomos, personal interno y desarrolladores). Cada historia incluye criterios de aceptación redactados en tercera persona y bajo la estructura Gherkin (Given-When-Then), garantizando que los requisitos sean claros, comprobables y agnósticos a la interfaz gráfica, los cuales permiten validar el cumplimiento de los requisitos de manera clara, comprobable y sin dependencia de detalles de interfaz de usuario. Adicionalmente, se incluyen *Technical Stories* orientadas a la implementación de componentes internos del sistema, como **APIs RESTful**, procesamiento en la nube e integración con dispositivos IoT.

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
EP-001	Digital Acquisition and Onboarding	Turn visitors into active users through a seamless sign-up experience	N/A (Es un Epic)	EP-001-US001, EP-001-US002, EP-001-US003, EP-001-US004, EP-001-US005, EP-001-US006, EP-001-US007, EP-001-TS008
EP-001-US001	Browse Landing Page Content	As a visitor, the user wants to browse landing page content so that the value proposition can be achieved.	Scenario 1: View value proposition Given that the visitor accesses the home page When the page loads completely Then the system displays the purpose, the key benefits, and an	EP-001

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			option to start a trial.	
			Scenario 2: View available plans Given that the visitor is on the home page When the visitor navigates to the plans section Then the system allows the visitor to compare options based on features, prices, and terms.	
EP-001-US002	Register as Lead	As a visitor, the user wants to register as a lead so that a business relationship can be established.	Scenario 1: Successful registration Given that the visitor fills out the form with valid information When the form is submitted Then the system confirms the account, requests email verification, and guides the user through the next steps.	EP-001
			Scenario 2: Registration with invalid data Given that the visitor enters incomplete or incorrectly formatted information When the form submission is attempted Then the system indicates which fields need to be corrected without losing the valid	

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			information already entered.	
EP-001-US003	Verify Email Account	As a newly registered user, the user wants to verify the email account so that identity can be confirmed.	Scenario 1: Successful verification Given that the system sends a valid verification link When the user accesses the link Then the system confirms the verification and activates the account. Scenario 2: Expired verification link Given that the verification link has expired When the system checks the validity Then the system notifies the user of the expiration and offers an option to request a new one.	EP-001
EP-001-US004	Complete Onboarding Wizard	As a new user, the user wants to complete the onboarding wizard so that plot data can be configured.	Scenario 1: Configure plot data and sensors Given that the user is in the initial setup flow When the user enters plot data and irrigation zones Then the system guides the user through the steps and allows saving progress to continue later. Scenario 2: Finalize onboarding Given that the user has	EP-001

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			completed all setup steps When the configuration is confirmed Then the system redirects to the dashboard and presents a guided tour of the functionalities.	
EP-001-US005	Browse Landing as Agronomist Visitor	As an agronomist visitor, the user wants to browse the landing page so that specific consultant benefits can be viewed.	Scenario 1: View the section for agronomists Given that the visitor accesses the landing page and selects the agronomist option When navigation to the corresponding section occurs Then the system displays benefits for consultants and a tailored registration form.	EP-001
EP-001-US006	Watch Product Demo Video	As a visitor, the user wants to watch the product demo video so that real-world use cases can be understood.	Scenario 1: Play embedded product video Given that the visitor is on the landing page When the video playback is triggered Then the system loads and plays the video correctly, showing real-world use cases, and prompts for sign-up at the end.	EP-001
EP-001-US007	Request Commercial Demo	As a visitor, the user wants to request a commercial demo so that a business	Scenario 1: Submit demo request form Given that the visitor fills out the demo	EP-001

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		relationship can be established.	request form When the information is submitted Then the system records the request, confirms that a representative will contact the visitor, and notifies the sales team. Scenario 2: Contact via WhatsApp from the landing page Given that the visitor prefers immediate contact When the WhatsApp contact action is triggered Then the system redirects to WhatsApp with a pre-filled message and logs the event for tracking.	
EP-002	Smart Monitoring and Water Management	Centralize soil health monitoring and automate irrigation to optimize resources	N/A (Es un Epic)	EP-002-US001, EP-002-US002, EP-002-US003, EP-002-US004, EP-002-US012, EP-002-US013, EP-002-US018, EP-002-TS013, EP-002-TS015, EP-002-TS005, EP-002-TS009, EP-002-TS026
EP-002-US001	View Real-Time Soil Dashboard	As an agriculturist, the user wants to view the real-time soil dashboard so that moisture, EC, pH, and	Scenario 1: View critical status Given that a soil parameter is out of the safe range When the dashboard is	EP-002

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EP-002-US002	Receive Irrigation Alert	As an agriculturist, the user wants to receive irrigation alerts so that immediate notification is generated when the soil reaches critical conditions.	temperature can be monitored at any time. accessed Then the system highlights the affected area and displays an initial recommendation. Scenario 2: View historical trends Given that historical data is required When the history is requested Then the system displays the parameters changes compared to the recommended ranges.	EP-002
			Scenario 1: Receive alert through preferred channel Given that the sensor reports critical conditions When the system evaluates the agronomic rules Then the system dispatches a notification through the selected channels, including the affected area and recommended actions.	
			Scenario 1: Activate irrigation successfully Given that a crop area requires irrigation When the activation is requested Then the system sends the command, confirms the start of the water	
EP-002-US003	Activate Irrigation Remotely	As an agriculturist, the user wants to activate irrigation remotely so that the field can be watered without physical presence.		EP-002

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			flow, and registers the action in the log.	
EP-002-US004	Stop Irrigation Manually	As an agriculturist, the user wants to stop irrigation manually so that watering can be interrupted early to avoid water waste.	Scenario 1: Stop irrigation early Given that the irrigation process is currently active When the stop action is requested Then the system sends the stop command, confirms the halt of the water flow, and estimates the saved water.	EP-002
EP-002-US012	View EC and pH Trends	As an agriculturist, the user wants to view EC and pH trends so that soil chemistry can be analyzed over time and salinity issues detected before they affect crops.	Scenario 1: View 7-day EC trend chart Given that the agriculturist is viewing a sensor's details When the EC trend chart is requested Then the system presents a line chart showing the last few days, reference lines, and alerts if the critical limit is exceeded. Scenario 2: Detect EC spike and show alert Given that the EC rises suddenly When the chart is evaluated Then the system highlights the increase and displays recommendations for correcting the salinity.	EP-002

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EP-002-US013	Receive High EC Alert	As an agriculturist, the user wants to receive a high EC alert so that soil salinity problems can be acted upon before they damage crops.	Scenario 1: Receive a high EC alert with recommendations Given that the EC exceeds the critical threshold When the system evaluates the telemetry data Then the system dispatches an alert detailing the issue, recent trends, and recommendations in plain language.	EP-002
			Scenario 2: EC alert combined with low moisture Given that EC is high and moisture is low simultaneously When both conditions are met during evaluation Then the system prioritizes the alert as Urgent and indicates that salt is blocking hydration.	
EP-002-US018	Calibrate Sensor Thresholds per Crop Type	As an agriculturist, the user wants to calibrate sensor thresholds per crop type so that alerts and recommendations are adjusted to the specific needs of the crops.	Scenario 1: Select crop type to auto-load thresholds Given that the agriculturist configures an irrigation zone When the crop type is selected Then the system automatically loads the recommended thresholds, allowing for	EP-002

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			manual adjustments. Scenario 2: Manually override a threshold with validation Given that a value outside the safe range is entered When the settings are saved Then the system issues a risk warning, requires confirmation, and logs the exception.	
EP-002-TS005	Cloud RESTful API - Telemetry Ingestion	As a Developer, the user wants the backend to implement a telemetry ingestion endpoint so that sensor data is securely received and stored.	Scenario 1: Bulk telemetry ingestion Given that the Edge node transmits a batch of sensor readings When the cloud backend receives the payload Then the system validates the data structure, persists the records, and returns a success confirmation.	EP-002
EP-002-TS006	Edge API Controls Irrigation Valve	As a Developer, the user wants the Edge component to process valve commands from the server so that irrigation is executed accurately (Server -> Embedded).	Scenario 1: Execute server irrigation command Given that the cloud server dispatches an open command When the Edge receives the instruction Then the Edge validates the request, activates the physical valve via the microcontroller, and confirms execution back to the cloud.	EP-002

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EP-002-TS007	Embedded ESP32 Reads Sensors	As a Developer, the user wants the firmware to periodically read sensors and transmit data to the Edge so that a continuous flow of soil data is guaranteed (Embedded -> Edge).	Scenario 1: Periodic reading cycle Given that the IoT device is powered on When the programmed reading cycle triggers Then the firmware reads the soil sensors, transmits the payload to the Edge, and returns to sleep mode.	EP-002
EP-002-TS008	Implement Biometric Auth for Valve Control	As a Developer, the user wants the mobile app to require biometric authentication for critical actions so that accidental valve activations are prevented.	Scenario 1: Require biometric confirmation Given that a user attempts to remotely open a valve When the activation trigger is pressed Then the system requests biometric validation before transmitting the command to the backend.	EP-002
EP-002-TS013	Cloud RESTful API — Parcels and Irrigation Zones	As a Developer, the user wants the backend to implement CRUD endpoints for parcels and irrigation zones so that the frontend can manage configuration securely.	Scenario 1: Create parcel with zones Given that the SPA sends new parcel data with defined zones When the backend processes the authenticated request Then the system creates the parcel, associates the zones with default thresholds, and returns the resource with its generated ID. Scenario 2: Update zone threshold	EP-002

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EP-002-TS015	Cloud RESTful API - Irrigation Commands	As a Developer, the user wants the backend to implement irrigation command endpoints so that the frontend can send valve orders with full traceability.	Given that a PATCH request is sent with new threshold limits When the backend validates the values Then the system updates the thresholds, logs the change in the audit trail, and returns the updated state.	EP-002
			Scenario 1: Queue irrigation command Given that the SPA or Mobile client sends a valve open order When the backend receives the command Then the system validates conflicts, queues it for Edge delivery, and returns a pending status.	
			Scenario 2: Update command status on Edge confirmation Given that the Edge node confirms execution When the backend receives the confirmation Then the system updates the status to executed and logs the timestamp.	
EP-002-TS009	Embedded ESP32 Controls Valve	As a Developer, the user wants the firmware to receive valve commands and execute them reliably so that irrigation is	Scenario 1: Auto-close valve when timer expires (safety fallback) Given that the valve is open and communication is	EP-002

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		activated and stopped correctly.	interrupted When the internal hardware timer reaches zero Then the device closes the valve automatically to prevent water waste.	
EP-002-TS026	Implement Biometric Auth for Critical Actions	As a Developer, the user wants the mobile app to require biometric authentication for critical actions so that accidental valve activations are prevented.	Scenario 1: Require biometric confirmation before valve activation Given that the user attempts to open an irrigation valve When the activation button is pressed Then the system requests fingerprint or Face ID and only proceeds if authentication is successful.	EP-002
EP-003	Perimeter Security and Early Warning	Protect farmers investments from intruders and adverse environmental conditions	N/A (Es un Epic)	EP-003-US005, EP-003-US006, EP-003-US007, EP-003-TS011, EP-003-TS006, EP-003-TS010, EP-003-SS001, EP-003-SS002
EP-003-US005	View Perimeter Security Events	As an agriculturist, the user wants to view perimeter security events so that intrusion attempts can be reviewed.	Scenario 1: View classified security events Given that the agriculturist accesses the security module When the system has recorded perimeter events Then the system presents a chronological	EP-003

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EP-003-US006	Receive Perimeter Intrusion Alert	As an agriculturist, the user wants to receive perimeter intrusion alerts so that human detections trigger real-time responses.	list showing the classification type (person/animal/wind) and confidence level.	EP-003
			Scenario 2: Filter events by type Given that the system contains multiple events When the agriculturist applies a specific classification filter Then the system displays only the events that match the requested filter.	
			Scenario 1: Receive critical alert for human detection Given that the system classifies an event as a human presence with high confidence When the classification is confirmed Then the system dispatches an immediate high-priority notification with location details. Scenario 2: Log low-priority alerts silently Given that the system classifies the event as an animal or wind When the evaluation occurs Then the system logs it in the history without dispatching a critical alert, unless explicitly configured.	

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EP-003-US007	Configure Perimeter Security Settings	As an agriculturist, the user wants to configure perimeter security settings so that detection sensitivity and active schedules can be defined.	Scenario 1: Adjust detection sensitivity Given that the agriculturist accesses the security settings When the sensitivity parameters are modified Then the system applies the settings to the physical device and confirms the changes.	EP-003
			Scenario 2: Define active monitoring schedule Given that the agriculturist configures a monitoring time window When the settings are saved Then the system ensures that events occurring outside of the scheduled times do not trigger active notifications.	
EP-003-TS011	Cloud RESTful API — Perimeter Security Endpoints	As a Developer, the user wants the backend to implement RESTful endpoints for perimeter security so that clients can query intrusion events.	Scenario 1: Retrieve paginated security events Given that a client application makes an authenticated request with valid filters When the system processes the request Then the backend returns a paginated JSON list of events with their respective classifications and timestamps.	EP-003

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EP-003-TS006	Edge API Processes PIR Classification Locally	As a Developer, the user wants the Edge to run the classification algorithm locally so that latency is reduced and offline operation is guaranteed.	Scenario 1: Classify perimeter event without cloud dependency Given that the Edge receives a motion trigger from the embedded device When the Edge processes the event with the local model Then it produces a classification (person/animal/wind) with a confidence score and stores it locally.	EP-003
			Scenario 2: Forward critical detection immediately Given that the classification results in 'Person' with high confidence When the Edge has active internet connectivity Then it dispatches the alert to the Cloud backend immediately.	
EP-003-TS010	Embedded ESP32 Processes PIR Motion Detection	As a Developer, the user wants the firmware to detect motion and capture raw data so that the Edge can classify the threat.	Scenario 1: Detect motion and capture classification data Given that the PIR sensor hardware detects a heat signature When the hardware interrupt is triggered Then the firmware captures the intensity and temporal pattern	EP-003

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			readings, and transmits the payload to the Edge.	
EP-003-SS001	Research Thermal PIR vs RGB Camera	As a Developer, the user wants to research and compare the accuracy and energy consumption of a thermal PIR sensor vs an RGB camera so that the best hardware for the security MVP can be defined.	Scenario 1: Execute field comparison tests Given that technical data is required to define the architecture When the team runs comparative field tests (person, animal, wind) Then a technical report is generated containing accuracy and energy consumption metrics to support the decision.	EP-003
EP-003-SS002	Research ML Model Size for Edge Classification	As a Developer, the user wants to research the maximum classification model size executable on the ESP32 so that the AI architecture can be defined (whether classification occurs on device or Edge).	Scenario 1: Benchmark model inference on hardware Given that candidate models of different sizes are available When the team runs inference benchmarks on the target hardware Then latency, RAM, and energy metrics are obtained to document the architectural recommendation.	EP-003
EP-004	Mobile Field Operations and IoT Device Management	Enable farmers to operate, monitor, and control their IoT devices from mobile under real-world field conditions	N/A (Es un Epic)	EP-004-US008, EP-004-US009, EP-004-US010, EP-004-US011, EP-004-US017, EP-004-US019, EP-004-TS023, EP-004-TS024, EP-004-TS025, EP-004-MS005
EP-004-	Use App in Offline Mode	As an agriculturist, the user wants to use the	Scenario 1: View cached sensor data	EP-004

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US008		app in offline mode so that cached sensor data can be viewed even without an internet connection.	offline Given that the mobile device loses internet connection When the application is accessed Then the system presents the most recent synchronized data along with a timestamp indicating the last update. Scenario 2: Automatic sync upon reconnection Given that the application is operating in offline mode When the internet connection is restored Then the system synchronizes data in the background and removes the offline state indicators.	
EP-004-US009	Receive and Interact with Push Notifications Natively	As an agriculturist, the user wants to receive and interact with push notifications natively so that critical alerts can be acted upon directly from the device's notification center.	Scenario 1: Receive actionable push notification Given that a critical system alert is generated When the notification reaches the mobile device Then the system renders a native alert containing quick-action buttons that execute the command without opening the app.	EP-004

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EP-004-US010	Open Irrigation Valve from Alert	As an agriculturist, the user wants to open the irrigation valve directly from an alert so that watering can be activated immediately.	Scenario 2: Request push notification permission Given that this is the first time the user opens the app When the user logs in Then the system explains the benefit and requests notification permission while respecting the user's choice.	EP-004
			Scenario 1: Activate valve from notification Given that a water stress notification is received When the valve activation action is triggered from the alert Then the system dispatches the open command, confirms execution, and updates the background state.	
			Scenario 1: View real-time statuses Given that the irrigation control module is accessed When the application communicates with the backend Then the system displays the state of each valve (open/closed), elapsed irrigation time, and estimated water consumption.	
EP-004-US011	View Live Valve Status from Mobile	As an agriculturist, the user wants to view the live valve and irrigation status from the mobile app so that the operational state of the field is known.		EP-004

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EP-004-US017	Register and Activate IoT Device	As an agriculturist, the user wants to register and activate an IoT device so that physical sensors can be linked to the digital account.	Scenario 1: Complete device registration Given that the user initiates the device setup wizard When the QR code is scanned and the configuration is submitted Then the system verifies the credentials, links the device to the account, and confirms the first telemetry reception.	EP-004
EP-004-US019	View Device Connectivity Status	As an agriculturist, the user wants to view the device connectivity status so that offline sensors can be identified proactively.	Scenario 1: Identify offline devices Given that the user accesses the device management list When the system retrieves the fleet status Then the interface clearly indicates whether each device is online or offline, along with the battery level and last known reading.	EP-004
EP-004-TS023	Configure Firebase Cloud Messaging for Push Notifications	As a Developer, the user wants to integrate FCM in the mobile app so that farmers receive reliable push notifications on Android even with the app closed.	Scenario 1: Register device token on app startup Given that the user logs in for the first time on a device When the app initializes Then the system requests notification permission, registers the FCM token, and associates it with the user profile in the backend.	EP-004

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Scenario 2: Receive push notification with app in background Given that the app is in the background or closed When FCM delivers a notification Then the OS displays it as a native notification with a title, body, and quick actions.				
EP-004-TS024	Implement Deep Linking from Notification to Valve Control Screen	As a Developer, the user wants to implement deep linking from notifications directly to the valve control screen so that the farmer can act on an alert with a single tap.	Scenario 1: Tap notification opens correct zone control screen Given that the farmer receives a water stress notification for a specific zone When the notification is tapped Then the application opens and navigates directly to that zone's control screen with soil status and action buttons visible. Scenario 2: Deep link works from locked screen Given that the device is locked and a critical notification arrives When the user swipes the quick action 'Activate Irrigation' Then the system requests biometric unlock and, if confirmed,	EP-004

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			executes the valve open command.	
EP-004-TS025	Implement Offline Command Queue	As a Developer, the user wants the app to manage an offline command queue so that requested operations are executed upon reconnection.	Scenario 1: Queue and dispatch actions Given that the device is offline When the user triggers an irrigation command Then the system stores the command locally and dispatches it automatically to the cloud once connectivity is restored, provided the safety timeframe is not exceeded.	EP-004
EP-004-MS005	Implement Zero-Touch Device Provisioning	As a Developer, the user wants the backend to support zero-touch provisioning so that hardware activation via QR scanning is secure and automated.	Scenario 1: Secure device onboarding Given that a new physical device requests activation When the system receives the factory certificate and QR pairing data Then the backend validates the integrity, provisions unique operational credentials, and registers the device in the active fleet.	EP-004
EP-005	IoT and Edge Infrastructure	Ensure the technical reliability of devices and connectivity in the field	N/A (Es un Epic)	EP-005-TS004, EP-005-TS007, EP-005-TS020, EP-005-MS001, EP-005-MS006, EP-005-MS008, EP-005-MS009, EP-005-MS010, EP-005-MS011, EP-005-MS013

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EP-005-TS004	Edge API Receives and Validates Sensor Telemetry	As a Developer, the user wants the Edge component to receive and store telemetry locally so that data reaches the system even with intermittent connectivity.	Scenario 1: Receive and store telemetry data Given that the IoT device transmits readings periodically When the Edge processes the payload Then the system validates physically possible ranges, stores the data locally, and confirms receipt to the device.	EP-005
EP-005-TS007	Edge DB Manages Local Storage Under Connectivity Constraints	As a Developer, the user wants the Edge database to manage storage with a defined retention policy so that memory overflow is avoided during offline periods.	Scenario 1: Apply data retention policy at storage threshold Given that the Edge has been offline for an extended period and storage exceeds the limit When the system evaluates the local policy Then the system retains critical events, discards old routine telemetry, and logs the purge action locally.	EP-005
EP-005-TS020	Implement Real-Time Data Synchronization via WebSocket	As a Developer, the user wants to implement data synchronization via WebSocket so that clients update automatically without polling the server.	Scenario 1: Receive and render live update Given that the SPA or Mobile client maintains an active WebSocket connection When the backend emits a telemetry update Then the client system receives the payload and reflects the new value	EP-005

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			on the dashboard without reloading.	
EP-005-MS001	Implement Async Event Processing Queue for IoT Telemetry	As a Developer, the user wants the system to implement an asynchronous message queue so that ingestion is decoupled from rule processing.	Scenario 1: Process telemetry burst without data loss Given that multiple Edge devices transmit telemetry simultaneously When the messages reach the cloud ingestion endpoint Then the system queues them in order, confirms persistence immediately, and processes them asynchronously in the background.	EP-005
EP-005-MS006	Implement Over-The-Air Firmware Update (OTA)	As a Developer, the user wants the system to support remote firmware updates so that security fixes are distributed without field visits.	Scenario 1: Device receives and applies firmware update Given that a new validated firmware version is available in the Cloud When the backend dispatches the update notification Then the IoT device downloads the binary, verifies the checksum, applies the update, and confirms the new version.	EP-005
EP-005-MS008	Implement Edge-to-Cloud Sync Protocol	As a Developer, the user wants to implement a robust Edge-to-Cloud synchronization protocol so that no data is lost during	Scenario 1: Sync completes with acknowledgment per batch Given that the Edge maintains pending local data	EP-005

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		network intermittency (Server -> Cloud).	When the internet connection is established Then the Edge dispatches data in batches, waits for cloud confirmation per batch, and clears the local buffer upon success.	
EP-005-MS009	Implement Edge API Authentication for IoT Devices	As a Developer, the user wants to implement mutual TLS authentication between IoT devices and the Edge so that only authorized nodes can communicate.	Scenario 1: Device authenticates successfully with client certificate Given that a provisioned device attempts to connect to the local Edge When the Edge validates the embedded client certificate Then the system establishes a secure mutual TLS channel and accepts the telemetry.	EP-005
EP-005-MS010	Implement Local Rule Engine on Edge	As a Developer, the user wants the Edge to evaluate threshold rules locally so that critical actions are guaranteed even without cloud connectivity.	Scenario 1: Evaluate rules offline Given that the Edge operates without a cloud connection When a new sensor reading violates a predefined threshold Then the local engine triggers the corresponding physical command and queues the event for later cloud synchronization.	EP-005
EP-005-MS011	Implement Edge Heartbeat and Health Reporting	As a Developer, the user wants the Edge to transmit periodic	Scenario 1: Alert triggered when heartbeat is missed	EP-005

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		heartbeats so that offline nodes are detected automatically.	Given that an Edge node fails to transmit a heartbeat within the expected timeframe When the cloud monitoring service detects the absence Then the system marks the node as offline and dispatches a technical support alert.	
EP-005-MS013	Implement IoT Gateway Auto-Reconnection Protocol	As a Developer, the user wants to implement an automatic reconnection protocol with a command queue so that solenoid valve activation is guaranteed even with network intermittency.	Scenario 1: Automatic recovery after network failure Given that the gateway loses network connectivity When the physical link is restored Then the system executes an exponential backoff routine, syncs the pending commands, and executes critical actions without saturating the server.	EP-005
EP-006	Subscriptions, Billing and Revenue Operations	Manage service monetization, the subscription lifecycle, and automatic billing processes	N/A (Es un Epic)	EP-006-US014, EP-006-US015, EP-006-US016, EP-006-TS017
EP-006-US014	View and Manage Subscription Plan	As an agriculturist, the user wants to view and manage the subscription plan so that the current tier, benefits, and upgrade options can be evaluated.	Scenario 1: View current subscription details Given that the user accesses the account settings When the subscription section is loaded Then the system displays the active plan,	EP-006

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EP-006-US015	Manage Billing and Payment Methods	As an agriculturist, the user wants to manage billing and payment methods so that payment history is accessible and credit cards can be updated.	price, covered hectares, next billing date, and included benefits.	EP-006
			Scenario 2: Upgrade from Basic to Pro plan Given that the user selects a higher-tier plan When the change is confirmed Then the system updates the plan, activates the new features immediately, and confirms the prorated billing adjustment.	
EP-006-US015	Manage Billing and Payment Methods	As an agriculturist, the user wants to manage billing and payment methods so that payment history is accessible and credit cards can be updated.	Scenario 1: View payment history Given that the user accesses the billing section When the history is requested Then the system presents a chronological list of payments with amounts, dates, and the option to download PDF receipts.	EP-006
			Scenario 2: Handle failed payment Given that an automatic payment fails due to insufficient funds When the system detects the error Then it notifies the user, prompts for a payment method update, and maintains the service in	

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			a temporary grace period.	
EP-006-US016	Cancel Subscription and Request Hardware Return	As an agriculturist, the user wants to cancel the subscription and request hardware return so that the offboarding process is handled securely.	Scenario 1: Initiate subscription cancellation Given that the user selects to cancel the active subscription When the cancellation is confirmed Then the system sets the deactivation date at the end of the billing cycle and asks for an optional exit reason. Scenario 2: Receive offboarding instructions Given that the cancellation has been processed successfully When the status updates Then the system dispatches an automated email containing account closure steps, the data retention policy, and hardware return instructions.	EP-006
EP-006-TS017	Cloud RESTful API — Subscriptions & Billing	As a Developer, the user wants the backend to implement subscription endpoints integrated with an external payment provider so that plans are managed automatically.	Scenario 1: Process successful subscription payment webhook Given that the external payment provider dispatches a successful payment webhook When the system validates the cryptographic signature	EP-006

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			Then the backend updates the subscription status to 'active', extends the billing date, and logs the transaction. Scenario 2: Handle failed payment and trigger grace period Given that the external provider notifies a failed payment When the system processes the webhook Then the backend activates the grace period flag, notifies the customer via the communication service, and queues automatic retries.	
EP-007	Software Quality and User Experience	Verify the robustness, security, and accessibility of the entire technology platform.	N/A (Es un Epic)	EP-007-TS021, EP-007-TS022, EP-007-TS027, EP-007-TS028, EP-007-MS002, EP-007-MS003, EP-007-MS004, EP-007-MS014
EP-007-TS021	Implement Lazy Loading and Code Splitting for Dashboard Modules	As a Developer, the user wants the system to implement lazy loading of dashboard modules so that the initial load time is reduced.	Scenario 1: Load only active module on initial render Given that the user accesses the dashboard When the application initializes Then the system loads only the active module, deferring inactive modules until navigation occurs.	EP-007

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
EP-007-TS027	Internationalization Support	As a Developer, the user wants the system to support internationalization so that the platform can be used comfortably without language barriers.	Scenario 2: Maintain acceptable performance Given that an agronomist with multiple parcels accesses the dashboard When the multi-parcel view loads Then the system renders the interface quickly using virtualization for parcels outside the viewport.	
			Scenario 1: Switch language preference Given that the platform supports multiple languages When the language preference is updated Then the system updates the entire application content to the selected language.	
			Scenario 2: Locale-appropriate formatting Given that a specific language and region are configured When the system displays dates, numbers, or amounts Then the format corresponds to the conventions of the selected region.	EP-007
EP-007-TS028	Accessibility Compliance	As a Developer, the user wants the system to comply with	Scenario 1: Screen reader compatibility Given that the user	EP-007

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
		accessibility standards so that users with disabilities can access all features without barriers.	accesses the platform with a screen reader When navigation occurs through the application Then the system identifies and permits operation of all interactive elements via text descriptions. Scenario 2: Non-color-dependent information Given that the system communicates critical states or alerts When the information is rendered Then it is conveyed through multiple indicators (text, patterns, shapes) and not by color alone.	
EP-007-MS002	Implement Auto-Scaling for Backend Services	As a Developer, the user wants the cloud infrastructure to configure horizontal auto-scaling for backend services so that performance is maintained during peak loads.	Scenario 1: Scale out when CPU exceeds threshold Given that the CPU usage exceeds the predefined threshold for a sustained period When the auto-scaling policy evaluates the metrics Then the system automatically provisions new instances and distributes incoming traffic. Scenario 2: Scale in during low-traffic periods	EP-007

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
EP-007-MS003	Implement Data Retention and Archiving Policy	As a Developer, the user wants the backend to implement a historical data retention policy so that storage costs are balanced with data availability.	Given that the CPU usage drops below the minimum threshold When the policy evaluation runs Then the system terminates excess instances gracefully without dropping active connections.	EP-007
			Scenario 1: Archive old telemetry data Given that telemetry readings exceed the active storage timeframe (12 months) When the scheduled archiving process executes Then the system moves the records to low-cost storage while maintaining searchable indexes.	
			Scenario 2: Purge data after absolute retention limit Given that archived data exceeds the absolute retention limit (5 years) When the purge policy evaluates the records Then the system deletes the data permanently and logs the deletion event in the audit trail.	
EP-007-MS004	Implement Automated Backup and Recovery	As a Developer, the user wants the cloud infrastructure to configure automatic	Scenario 1: Daily backup completes successfully Given that the backup	EP-007

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
		backups of the main database so that data can be recovered after a failure.	process runs on schedule When the backup completes Then the system verifies integrity, stores it in a secure geographic region, and logs the event. Scenario 2: Validate recovery from backup Given that a periodic recovery test is executed When the backup is restored in a test environment Then the system ensures data is intact and the documented recovery time objective is met.	
EP-007-MS014	Setup CI/CD Pipeline for Modular Backend	As a Developer, the user wants the repository to configure a CI/CD pipeline so that code compilation, testing, and deployment are automated safely.	Scenario 1: Automated build and test on code integration Given that a code modification is integrated into the main branch When the pipeline is triggered Then the system executes compilation, unit tests, and integration tests automatically. Scenario 2: Deployment blocking on test failure Given that the	EP-007

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			automated test suite returns a failure When the deployment step is evaluated Then the pipeline halts the execution, prevents the deployment to production, and notifies the development team.	
EP-007-TS022	Implement Global State Management for Dashboard	As a Developer, the user wants to implement a global state manager on the web frontend so that sensor data, alerts, and configuration are synchronized between components without redundant API calls.	Scenario 1: Synchronize state across SPA components Given that a telemetry update is received via WebSocket or API When the global state store is updated Then all active UI components reflecting that data re-render automatically without making additional HTTP requests.	EP-007
EP-007-TS027	Internationalization Support	As a Developer, the user wants the system to support internationalization so that the platform can be used comfortably without language barriers.	Scenario 1: Switch language preference Given that the platform supports multiple languages When the language preference is updated Then the system updates the entire application content to the selected language. Scenario 2: Locale-appropriate formatting Given that a specific language and region are configured	EP-007

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
EP-007-TS028	Accessibility Compliance	As a Developer, the user wants the system to comply with accessibility standards so that users with disabilities can access all features without barriers.	When the system displays dates, numbers, or amounts Then the format corresponds to the conventions of the selected region.	EP-007
			Scenario 1: Screen reader compatibility Given that the user accesses the platform with a screen reader When navigation occurs through the application Then the system identifies and permits operation of all interactive elements via text descriptions.	
			Scenario 2: Non-color-dependent information Given that the system communicates critical states or alerts When the information is rendered Then it is conveyed through multiple indicators (text, patterns, shapes) and not by color alone.	
EP-007-MS002	Implement Auto-Scaling for Backend Services	As a Developer, the user wants the cloud infrastructure to configure horizontal auto-scaling for backend services so that performance is maintained during peak	Scenario 1: Scale out when CPU exceeds threshold Given that the CPU usage exceeds 70% for a sustained period When the auto-scaling policy evaluates the metrics	EP-007

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
		loads without unnecessary costs.	Then the system automatically provisions new instances and distributes incoming traffic. Scenario 2: Scale in during low-traffic periods Given that the CPU usage drops below 30% When the policy evaluation runs Then the system terminates excess instances gracefully without dropping active connections.	
EP-007-MS003	Implement Data Retention and Archiving Policy	As a Developer, the user wants to implement a historical data retention and archiving policy so that storage costs are balanced with data availability for analysis.	Scenario 1: Archive old telemetry data Given that telemetry readings exceed the active storage timeframe (12 months) When the scheduled archiving process executes Then the system moves the records to low-cost storage while maintaining searchable indexes. Scenario 2: Purge data after absolute retention limit Given that archived data exceeds the absolute retention limit (5 years) When the purge policy evaluates the records	EP-007

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
EP-007-MS004	Implement Automated Backup and Recovery	As a Developer, the user wants the cloud infrastructure to configure automatic backups of the main database so that data can be recovered after a catastrophic failure.	Then the system deletes the data permanently and logs the deletion event in the audit trail.	EP-007
			Scenario 1: Daily backup completes successfully Given that the backup process runs on schedule When the backup completes Then the system verifies integrity, stores it in a secure geographic region, and logs the event. Scenario 2: Validate recovery from backup Given that a periodic recovery test is executed When the backup is restored in a test environment Then the system ensures data is intact and the documented recovery time objective is met.	
EP-007-MS014	Setup CI/CD Pipeline for Modular Backend	As a Developer, the user wants the repository to configure a CI/CD pipeline so that code compilation, testing, and deployment of backend modules are automated safely.	Scenario 1: Automated build and test on code integration Given that a code modification is integrated into the main branch When the pipeline is triggered Then the system executes compilation,	EP-007

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			unit tests, and integration tests automatically. Scenario 2: Deployment blocking on test failure Given that the automated test suite returns a failure When the deployment step is evaluated Then the pipeline halts the execution, prevents deployment to production, and notifies the development team.	
EP-008	Notifications and Multi-Channel Communication	Deliver timely, reliable, and personalized alerts across Push, WhatsApp, and SMS to keep all users informed and able to act.	N/A (Es un Epic)	EP-008-US020, EP-008-US021, EP-008-TS018
EP-008-US020	Manage Advanced Notification Preferences	As an agriculturist, the user wants to manage advanced notification preferences so that alerts are routed only to the appropriate channels.	Scenario 1: Configure notification channels by alert type Given that the user accesses the notification settings When the channels are configured by alert type Then the system saves the preferences and routes future alerts strictly through the selected channels. Scenario 2: Receive daily summary digest Given that the user has opted in for a daily summary	EP-008

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
EP-008-US021	Receive WhatsApp Alert with Direct Action Link	As an agriculturist, the user wants to receive WhatsApp alerts with direct action links so that critical conditions can be addressed immediately from a messaging app.	When the scheduled system time arrives Then the system dispatches a consolidated message containing soil conditions, irrigation activities, and security events.	EP-008
			Scenario 1: Receive WhatsApp alert with deep link Given that a critical water stress alert is generated When the system dispatches the notification via WhatsApp Then the message includes a plain-language summary and a secure link that routes the user to the control screen.	
			Scenario 2: Alert escalates to SMS if WhatsApp is undelivered Given that a WhatsApp message is not confirmed as delivered When the configured timeout period elapses Then the system automatically escalates the alert by sending a backup SMS.	
EP-008-TS018	Cloud RESTful API — Notifications Dispatch	As a Developer, the user wants the backend to implement a multi-	Scenario 1: Dispatch queued notification to external provider	EP-008

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
		channel notification dispatch service so that internal modules can queue alerts for external delivery (Cloud).	Given that an internal module queues a critical notification When the dispatch service processes the queue Then the system identifies the preferred channel and executes the API call to the external provider (e.g., Twilio). Scenario 2: Handle external provider failure Given that the external communication provider returns an error response When the dispatch is attempted Then the system logs the failure for analysis and requeues the message using exponential backoff.	
EP-009	Agronomist Consulting and Multi-Client Management	Empower agronomists to remotely supervise multiple client parcels, standardize recommendations, and deliver data-driven consulting at scale.	N/A (Es un Epic)	EP-009-US001, EP-009-US002, EP-009-US003, EP-009-US004, EP-009-US005, EP-009-TS016, EP-009-MS012
EP-009-US001	View Multi-Parcel Dashboard	As an agronomist, the user wants to view a multi-parcel dashboard so that all clients' parcels can be monitored from a single consolidated view.	Scenario 1: View an overview of all client parcels Given that the agronomist has multiple active clients When the main dashboard is accessed	EP-009

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			Then the system displays a consolidated view with status indicators for every associated parcel. Scenario 2: Access quick parcel details Given that the agronomist is viewing the parcel list When quick information about a specific parcel is requested Then the system presents key metrics and alert statuses without requiring navigation to the full details view.	
EP-009-US002	Adjust Client Irrigation Thresholds Remotely	As an agronomist, the user wants to adjust client irrigation thresholds remotely so that crop recommendations can be updated without a physical field visit.	Scenario 1: Update thresholds for a single parcel Given that the agronomist manages a specific parcel's settings When the threshold values are modified and confirmed Then the system applies the new values, notifies the associated farmer, and logs the change. Scenario 2: Bulk update for multiple parcels Given that the agronomist selects multiple similar parcels When a threshold change is applied	EP-009

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			globally Then the system processes the update for each parcel simultaneously and confirms the operation.	
EP-009-US003	Generate Technical Report for Client	As an agronomist, the user wants to generate a technical report so that monthly soil health summaries and recommendations can be shared in a professional format.	Scenario 1: Generate monthly soil health report Given that the agronomist requires a report for a specific plot When the report generation is requested Then the system compiles sensor data, historical trends, and applied recommendations into an exportable document. Scenario 2: Share report directly with client Given that a report has been generated successfully When the external share action is triggered Then the system prepares a direct delivery method containing a summary and a secure download link.	EP-009
EP-009-US004	Monitor Remote Parcels Between Visits	As an agronomist, the user wants to monitor remote parcels between visits so that proactive recommendations can be dispatched upon	Scenario 1: Receive critical alert for a client's plot Given that a client's sensor detects a critical telemetry value	EP-009

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
		critical environmental changes.	When the evaluation rules are triggered Then the system dispatches an alert to the assigned agronomist detailing the parameter and historical context. Scenario 2: Dispatch recommendation from alert context Given that the agronomist evaluates an incoming critical alert When a professional recommendation is formulated Then the system routes the message directly to the farmer accompanied by the supporting sensor data.	
EP-009-US005	Configure Crop-Specific Thresholds as Agronomist Template	As an agronomist, the user wants to configure crop-specific thresholds as templates so that standardized settings can be quickly applied to new client parcels.	Scenario 1: Create a custom threshold template Given that the agronomist accesses the template module When thresholds are defined and saved as a template Then the system associates it with the professional's account for future use. Scenario 2: Apply template to a client parcel Given that a saved template exists When the agronomist	EP-009

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			applies it to a specific client parcel Then the system overwrites the current thresholds, notifies the farmer, and records the action in the audit trail.	
EP-009-TS016	Cloud RESTful API — Agronomic Diagnosis Engine	As a Developer, the user wants the backend to implement an agronomic diagnosis endpoint combining soil data with weather context so that automatic recommendations are generated.	Scenario 1: Generate diagnosis on demand Given that a client application requests a diagnosis for a specific zone When the backend combines soil readings with meteorological data Then the system returns a computed water stress index and a structured recommendation. Scenario 2: Trigger automatic diagnosis when threshold is crossed Given that a telemetry reading exceeds a critical threshold When the rules engine evaluates the condition Then the system generates a diagnosis automatically and queues a notification for the assigned agronomist.	EP-009
EP-009-MS012	Integrate External Weather API to Enrich Diagnosis	As a Developer, the user wants the system to integrate an external weather API so that the diagnosis engine considers climate	Scenario 1: Enrich diagnosis with weather context Given that the rules engine evaluates irrigation conditions	EP-009

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
		conditions when calculating water stress.	When external weather data (e.g., imminent rain forecast) is retrieved Then the system adjusts the water stress calculation to prevent over-irrigation. Scenario 2: Graceful degradation when external service is unavailable Given that the external weather service is unresponsive When the system requires climate data Then it proceeds using only internal soil telemetry and flags the diagnosis as lacking external weather context.	
EP-010	Customer Support and IoT Fleet Monitoring	Provide support staff with tools to monitor device health fleet-wide, detect incidents, and resolve issues without field visits.	N/A (Es un Epic)	EP-010-US001, EP-010-TS029, EP-010-MS007
EP-010-US001	Monitor IoT Device Health Fleet	As a support staff member, the user wants to monitor the IoT device health fleet so that deployed devices can be managed and maintained proactively without unnecessary field visits.	Scenario 1: View device health dashboard Given that the support staff accesses the device management module When the fleet status is requested Then the system presents the connection, battery, and signal status for each deployed device, allowing filtering	EP-010

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			by critical issues.	
			Scenario 2: Schedule maintenance for devices with low battery Given that the system identifies devices with critically low batteries When a maintenance visit is scheduled Then the system generates a work order, notifies the assigned technician, and dispatches a notification to the associated farmer with the date.	
EP-010-TS029	Cloud RESTful API — Fleet Health Endpoints	As a Developer, the user wants the backend to implement fleet health endpoints so that the support SPA can retrieve and aggregate connection and battery statuses securely (Cloud).	Scenario 1: Retrieve aggregated fleet status Given that the support application requests the global fleet status When the backend receives the authenticated request Then the system queries the active devices, aggregates the health metrics, and returns a paginated JSON response. Scenario 2: Handle unauthorized access attempt Given that a client without support privileges attempts to access the fleet endpoint When the request is	EP-010

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			evaluated Then the system denies access and returns a 403 Forbidden status.	
EP-010-MS007	Implement Remote Hardware Diagnostics	As a Developer, the user wants to implement remote hardware diagnostics so that the support system can execute tests on the embedded sensors and valves directly from the cloud (Cloud -> Embedded).	Scenario 1: Request remote diagnostic from the cloud Given that a farmer reports an issue with their physical device When the support staff triggers a remote diagnostic command Then the backend dispatches the instruction, the firmware executes tests on the hardware components, and returns a status report. Scenario 2: Identify faulty sensor via diagnostic report Given that the diagnostic report is received by the backend When the system processes the results Then it identifies faulty components, generates an action recommendation, and logs the result in the customer's support ticket.	EP-010
EP-011	Ops, Admin and Customer Account Management	Enable the Ops & Billing team to manage customer lifecycle, device activations, account suspensions, and internal	N/A (Es un Epic)	EP-011-US001, EP-011-US002, EP-011-TS030

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
		administrative workflows.		
EP-011-US001	Manage Customer Accounts from Admin Panel	As an operations specialist, the user wants to manage customer accounts from the admin panel so that the customer lifecycle and account suspensions can be controlled.	Scenario 1: Search for and view a customer account Given that the operations specialist accesses the customer management module When a specific customer search is executed Then the system presents the full profile, including the active plan, linked devices, payment history, and account status. Scenario 2: Suspend account for payment default Given that a customer has exceeded the payment grace period When the suspension command is executed Then the system disables the user's platform access, retains the historical data, dispatches a notification to the customer, and logs the administrative action. Scenario 3: Reactivate suspended account Given that a suspended customer has settled their payment When the account	EP-011

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			reactivation is triggered Then the system restores full platform access, resumes device telemetry synchronization, and notifies the customer.	
EP-011-US002	Manage and Activate IoT Devices from Admin Panel	As an operations specialist, the user wants to manage and activate IoT devices from the admin panel so that new hardware batches are registered and compromised devices are blocked.	Scenario 1: Register a new batch of devices Given that the operations team receives a new hardware batch When a file containing IDs and factory serial numbers is uploaded Then the system registers the devices in the database and updates their status to 'available for activation'. Scenario 2: Deactivate a device reported as lost Given that a customer reports a physical device as lost or stolen When the deactivation action is performed by the specialist Then the system invalidates the device's security credentials, permanently blocks its telemetry ingestion, and records the deactivation reason.	EP-011
EP-011-TS030	Cloud RESTful API — Admin Operations Endpoints	As a Developer, the user wants the backend to implement administrative	Scenario 1: Process batch device registration payload Given that the admin	EP-011

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
		endpoints so that the back-office SPA can securely manage accounts and device registries (Cloud).	SPA transmits a batch registration payload When the backend receives the authenticated request Then the system validates the administrative privileges, parses the serial numbers, persists the new registry, and returns a success summary. Scenario 2: Secure suspension endpoint execution Given that an account suspension request is received When the system processes the endpoint call Then the backend updates the account flag to 'suspended', invalidates active user sessions (JWT), and queues the suspension notification.	
EP-012	Business Intelligence and Strategic Analytics	Provide strategic dashboards and feature adoption metrics to allow the Product Owner to make data-driven decisions and monitor business goals.	N/A (Es un Epic)	EP-012-US022, EP-012-US023, EP-012-TS031
EP-012-US022	View Business Strategic Analytics	As a product owner, the user wants to review growth, retention, and churn metrics by segment so	Scenario 1: View executive dashboard with KPIs Given that the product owner accesses the	EP-012

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
		that the quarterly roadmap and strategies can be adjusted.	executive dashboard When the quarter's metrics are reviewed Then the system presents Active Users, MRR, Conversion Rate, and Churn with comparisons to the previous period. Scenario 2: Analyze churn by segment Given that the product owner analyzes cancellation reasons When the segment filter is applied Then the system displays churn statistics by customer type and the main reported causes.	
EP-012-US023	View Feature Adoption and Usage Heatmap	As a product owner, the user wants to see which features are used and how frequently by segment so that roadmap improvements can be prioritized based on real usage.	Scenario 1: View feature usage by user segment Given that the product owner accesses the product analytics module When a specific period and segment are selected Then the system displays the usage frequency of each feature, identifying the most and least used. Scenario 2: Identify features with high abandonment rate Given that the product	EP-012

Epic / Story ID	Título	Descripción	Criterios de Aceptación	Relacionado con (Epic ID)
			owner analyzes the funnel of a specific feature When the flow steps are reviewed Then the system identifies the step where users drop off most frequently and how many complete the action.	
EP-012-TS031	Cloud RESTful API — Business Intelligence Endpoints	As a Developer, the user wants the backend to implement analytical endpoints so that the BI SPA can retrieve aggregated usage and financial metrics (Cloud).	Scenario 1: Aggregate churn and adoption metrics Given that the BI SPA requests the quarterly analytics payload When the backend processes the request Then the system queries the operational and billing databases, aggregates the data by segment, and returns the computed metrics securely. Scenario 2: Secure analytics access Given that an unauthorized user attempts to access the BI endpoints When the request is evaluated Then the system denies access, returns a 403 Forbidden status, and logs the security event.	EP-012

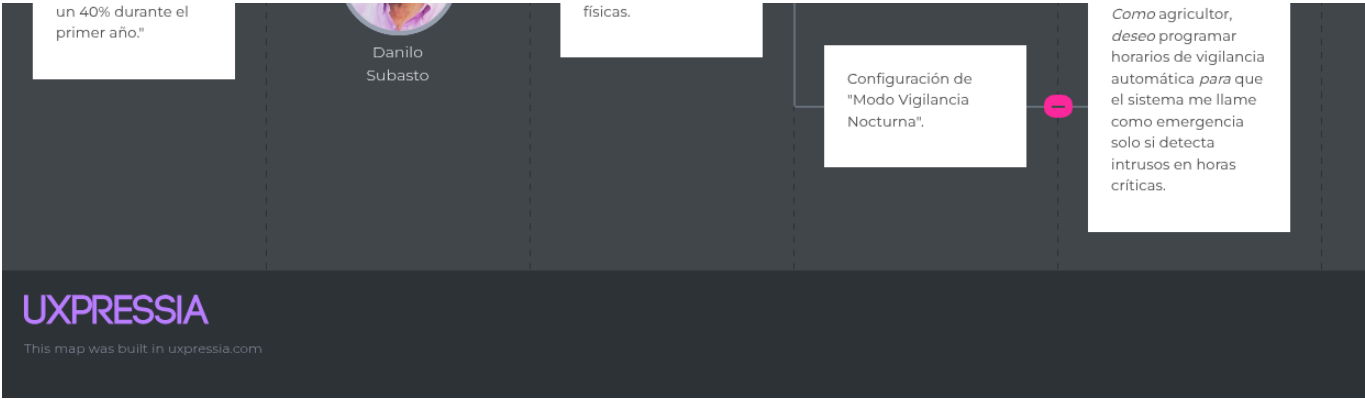
3.2. Impact Mapping

En esta sección, presentamos el **Impact Mapping** el cual como finalidad alinear los objetivos de negocio que tiene SATECHO, bajo la metodología **SMART** (Specific, Measurable, Attainable, Relevant y Timely), con las necesidades y comportamientos esperados de cada uno de nuestros **User Personas**. A través de esta actividad, identificamos cómo los actores clave (Agricultor e Ingeniero Agrónomo) deben cambiar su comportamiento (Impacts) para ayudarnos a alcanzar las metas, qué funcionalidades o artefactos construiremos (Deliverables) para provocar esos cambios, y cómo se traducen estos en historias de usuario (User Stories) concretas para el equipo de desarrollo.

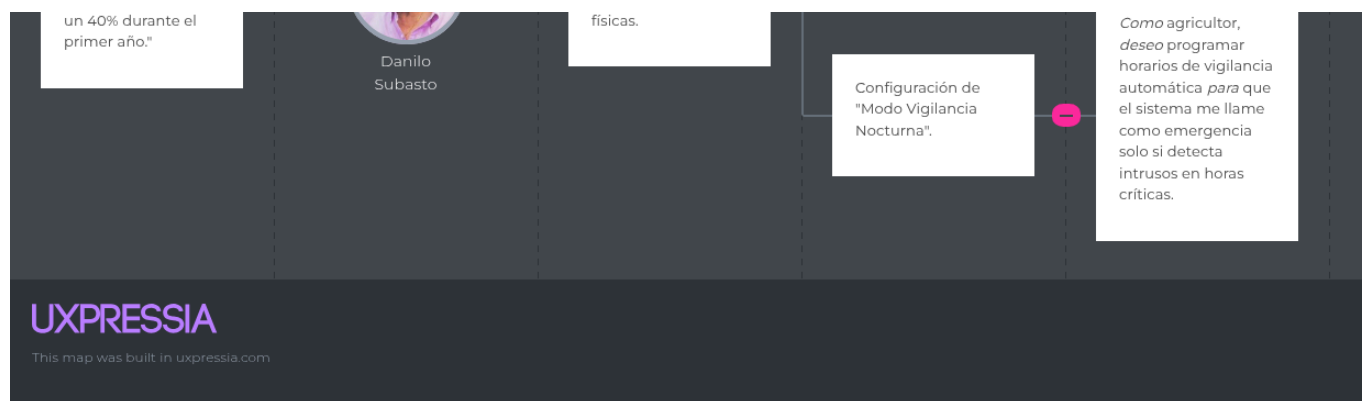
Agricultor

El Impact Mapping asociado a Danilo Subasto representa la alineación estratégica entre los objetivos de negocio planteados para la solución digital orientada al sector agrícola, las necesidades reales del usuario final y los entregables funcionales del producto de software que permitirán generar valor medible. Este mapa de impacto permite visualizar cómo cada iniciativa tecnológica responde directamente a los retos operativos del agricultor, asegurando que el desarrollo de la plataforma no se enfoque únicamente en funcionalidades aisladas, sino en resultados concretos que impulsen la adopción, eficiencia operativa y seguridad del usuario.









En primer lugar, se establece como objetivo comercial la adquisición de 50 agricultores suscritos al plan Premium durante los primeros seis meses de operación. Para alcanzar esta meta, se identifica que Danilo Subasto necesita confiar en la solución antes de realizar una inversión económica, por lo que debe percibir beneficios claros, casos reales de éxito y una propuesta de valor transparente. En respuesta a ello, el producto contempla entregables clave como una landing page informativa, donde se explique el servicio de manera clara y se presenten testimonios verificables de otros productores; un formulario rápido para solicitar pilotos gratuitos, que reduzca barreras de entrada y facilite la prueba del sistema; y una página de precios estructurada, donde se comparen los planes Básico y Premium de forma sencilla. Estas funcionalidades buscan convertir el interés inicial en registros efectivos y futuras suscripciones pagadas.

Como segundo eje estratégico, se plantea lograr que el 80% de los usuarios activos reduzcan su consumo de agua en al menos un 20% durante su primer ciclo de cultivo, promoviendo eficiencia hídrica y sostenibilidad productiva. Desde la perspectiva del usuario, Danilo necesita conocer con precisión el estado de humedad de su terreno para dejar de regar por intuición o por costumbre. Para ello, la solución propone un dashboard web y móvil en tiempo real, con una interfaz visual tipo semáforo (verde, amarillo y rojo) que permita interpretar rápidamente la condición del suelo sin requerir conocimientos técnicos avanzados. Complementariamente, se integra un sistema de alertas automáticas por SMS o WhatsApp que notifique episodios de estrés hídrico y un módulo de control remoto para electroválvulas, permitiendo activar o detener el riego desde la aplicación. Estas capacidades tecnológicas facilitan decisiones oportunas, ahorro de recursos y automatización operativa.

Finalmente, el tercer objetivo de negocio consiste en reducir en un 40% los reportes de incidentes de seguridad y falsas alarmas durante el primer año. En este contexto, Danilo requiere disminuir la necesidad de vigilancia nocturna presencial y contar con mecanismos inteligentes que filtren eventos irrelevantes. Para responder a esta necesidad, se plantea un módulo de seguridad con clasificación de amenazas mediante Inteligencia Artificial, capaz de diferenciar entre personas, animales u otros movimientos no críticos, reduciendo alertas innecesarias. Asimismo, se incorpora una configuración de Modo Vigilancia Nocturna, mediante la cual el usuario puede programar horarios específicos para activar monitoreo reforzado únicamente en franjas horarias sensibles. Estas funcionalidades incrementan la tranquilidad del agricultor, optimizan tiempos de supervisión y fortalecen la protección de la infraestructura agrícola.

Ingeniero Agrónomo

El Impact Mapping asociado a Nestor Campo representa la alineación estratégica entre nuestros objetivos de negocio y las necesidades reales de los ingenieros agrónomos independientes que interactuarán con el sistema. Este mapa visualiza de manera estructurada cómo el comportamiento de nuestro usuario clave impacta directamente en el éxito de la plataforma, y detalla qué entregables de software son necesarios para facilitar dicho comportamiento, garantizando que todo el esfuerzo de desarrollo esté justificado por el valor que aporta.









En cuanto al primer eje estratégico, el cual se titula **Adquisición y Adopción Inicial de la Plataforma**, la estrategia se centra en la transición del agrónomo desde métodos tradicionales hacia un ecosistema digital centralizado. Para alcanzar la meta de 20 profesionales registrados y la vinculación efectiva de clientes en el primer semestre, el producto despliega una Landing Page B2B diseñada específicamente para asesores técnicos y cooperativas, actuando como el primer punto de contacto donde se comunica el valor de la escalabilidad operativa. Este proceso se consolida mediante un portal de Onboarding seguro que no solo gestiona la identidad del usuario, sino que le otorga la capacidad técnica de administrar múltiples perfiles de clientes desde una única cuenta centralizada, reduciendo significativamente la fricción de entrada y los costos de desplazamiento físico desde el primer día.

Respecto al segundo eje llamado **Eficiencia Operativa y Diagnóstico Remoto**, la plataforma busca escalar la capacidad de gestión del ingeniero en un 50% mediante la implementación de herramientas de monitoreo de alta precisión. El núcleo técnico de esta solución reside en un Dashboard web centralizado con vista multi-cliente, el cual permite la supervisión simultánea de diversas parcelas e identifica anomalías críticas mediante un sistema de alertas visuales. Este diagnóstico se sustenta en un módulo de gráficos históricos avanzados que procesan variables críticas como humedad, conductividad eléctrica (EC) y pH, permitiendo un análisis científico de tendencias. Al integrar paneles de configuración de umbrales personalizados, el sistema automatiza la detección de estrés hídrico o salinidad, notificando al usuario en tiempo real cuando los niveles se desvían del rango óptimo, lo que transforma la consultoría presencial en una gestión basada en datos y respuesta inmediata.

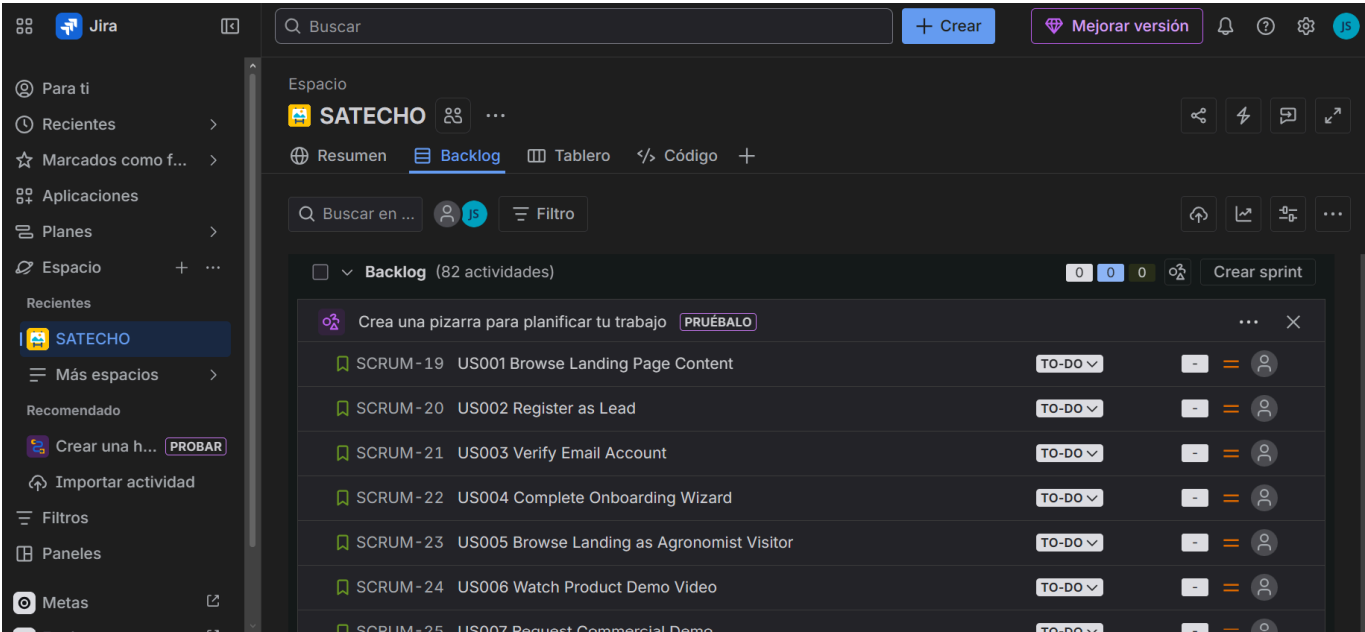
Finalmente, como último eje asociado al ámbito de la **Automatización de Tareas y Profesionalización del Servicio**; el enfoque se dirige a la sustitución de registros manuales y hojas de cálculo por flujos de trabajo automatizados que aseguren un cumplimiento del 90% en la entrega de informes técnicos. La plataforma integra un generador automático de reportes PDF que sintetiza los datos agronómicos recolectados durante la semana en un entregable profesional listo para ser distribuido a través de canales de mensajería instantánea. Como complemento crítico para la gobernanza del riego, se incluye un registro de actividad (log) que monitorea el estado de las electroválvulas. Esta funcionalidad permite al agrónomo realizar una auditoría técnica sobre la ejecución de sus recomendaciones en el campo, verificando tiempos exactos de activación de bombas de agua y garantizando que el agricultor cumpla rigurosamente con las instrucciones de riego proporcionadas, elevando así el estándar de calidad del servicio de asesoría.

3.3. Product Backlog

En esta sección presentamos el motor estratégico de Satecho: nuestro **Product Backlog**. Más que un inventario de requerimientos gestionado a través de Jira, concebimos este espacio como una hoja de ruta viva

y enfocada en la entrega de valor continuo. Para asegurar una planificación transparente y realista, hemos dimensionado cada historia de usuario utilizando la escala de Fibonacci (1, 2, 3, 5, 8). Esta técnica ágil nos permite estimar el esfuerzo y la complejidad de forma relativa, facilitando que tanto el equipo de desarrollo como los tomadores de decisiones compartan una misma perspectiva sobre lo que implica cada reto. En consecuencia, la estructura de nuestro backlog no es aleatoria; está rigurosamente priorizada de forma descendente.

En la cúspide se posicionan las iniciativas de mayor impacto y criticidad para los objetivos del negocio, garantizando que nuestros recursos se dirijan siempre a construir primero lo que Satecho necesita con mayor urgencia.



Podrá observar el *Product Backlog*, desarrollado en Jira en el enlace que se presenta a continuación: [Product Backlog-Satecho](#)

El *Product Backlog* detallado lo puede observa en la siguiente tabla:

# Orden	User Story Id	Título	Descripción	Story Points (1 / 2 / 3 / 5 / 8)
1	EP-001-US001	Browse Landing Page Content	As a visitor, I want to browse landing page content so that I can understand the product's value proposition and available plans.	2
2	EP-001-US005	Browse Landing as Agronomist Visitor	As an agronomist visitor, I want to browse the landing page so that I can see the benefits tailored to consultants.	1
3	EP-001-US006	Watch Product Demo Video	As a visitor, I want to watch the product demo video so that I can understand real-world use cases before signing up.	1

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5	EP-001-US002	Register as Lead	As a visitor, I want to register as a lead so that I can create my account and start the onboarding process.	3
6	EP-001-US003	Verify Email Account	As a newly registered user, I want to verify my email account so that I can confirm my identity and activate my access.	3
7	EP-001-US004	Complete Onboarding Wizard	As a newly registered user, I want to complete the onboarding wizard so that I can configure my plot data, irrigation zones, and get guided into the platform from day one.	5
8	EP-002-US001	View Real-Time Soil Dashboard	As an agriculturist, I want to view the real-time soil dashboard so that I can monitor moisture, EC, pH, and temperature of my parcels at any time.	8
9	EP-002-US012	View EC and pH Trends	As an agriculturist, I want to view EC and pH trends so that I can analyze soil chemistry over time and detect salinity issues before they affect my crops.	5
10	EP-002-US002	Receive Irrigation Alert	As an agriculturist, I want to receive irrigation alerts so that I am notified immediately when the soil reaches critical conditions and I can take action.	5
11	EP-002-US003	Activate Irrigation Remotely	As an agriculturist, I want to activate irrigation remotely so that I can start watering my field from anywhere without being physically present.	8
12	EP-002-US004	Stop Irrigation Manually	As an agriculturist, I want to stop irrigation manually so that I can interrupt watering early and avoid water waste when conditions change.	3
13	EP-002-TS007	Embedded ESP32 Reads Sensors	As a Developer, I want the firmware to periodically read sensors and transmit data	5

# Orden	User Story Id	Título	Descripción	Story Points (1 / 2 / 3 / 5 / 8)
			to the Edge so that a continuous flow of soil data is guaranteed.	
14	EP-005-TS004	Edge API Receives and Validates Sensor Telemetry	As a Developer, I want the Edge component to receive and store telemetry locally so that data reaches the system even with intermittent connectivity.	8
15	EP-002-TS005	Cloud RESTful API - Telemetry Ingestion	As a Developer, I want the backend to implement a telemetry ingestion endpoint so that sensor data is securely received and stored.	5
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20	EP-004-TS025	Implement Offline Command Queue	As a Developer, I want to manage an offline command queue in the mobile app so that requested irrigation operations are executed automatically upon reconnection.	8
21	EP-004-US009	Receive and Interact with Push Notifications Natively	As an agriculturist, I want to receive and interact with push notifications natively so that I can be alerted of critical conditions directly from my device's notification center.	3
22	EP-004-TS023	Configure Firebase Cloud Messaging for Push	As a Developer, I want to integrate FCM in the mobile app so that farmers receive	5

# Orden	User Story Id	Título	Descripción	Story Points (1 / 2 / 3 / 5 / 8)
		Notifications	reliable push notifications on Android even with the app closed.	
23	EP-004-TS024	Implement Deep Linking from Notification to Valve Control Screen	As a Developer, I want to implement deep linking from notifications directly to the valve control screen so that the farmer can navigate to the exact plot with a single tap.	5
24	EP-004-US010	Open Irrigation Valve from Alert	As an agriculturist, I want to open the irrigation valve directly from an alert so that I can activate watering immediately without having to manually search for the parcel in the app.	3
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27	EP-003-TS010	Embedded ESP32 Processes PIR Motion Detection	As a Developer, I want the firmware to detect motion and capture raw data so that the Edge can classify the threat.	5
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30	EP-003-US005	View Perimeter Security Events	As an agriculturist, I want to view perimeter security events so that I can review a classified log of intrusion attempts and filter by event type.	5
31	EP-003-US006	Receive Perimeter Intrusion Alert	As an agriculturist, I want to receive perimeter intrusion alerts so that I can	3

# Orden	User Story Id	Título	Descripción	Story Points (1 / 2 / 3 / 5 / 8)
			respond to human detections in real-time.	
32	EP-003-US007	Configure Perimeter Security Settings	As an agriculturist, I want to configure perimeter security settings so that I can adjust detection sensitivity and define active monitoring schedules.	5
33	EP-009-US001	View Multi-Parcel Dashboard	As an agronomist, I want to view a multi-parcel dashboard so that I can monitor all my clients' parcels from a single consolidated view.	8
34	EP-009-TS016	Cloud RESTful API — Agronomic Diagnosis Engine	As a Developer, I want to implement an agronomic diagnosis endpoint combining soil data with weather context so that automatic recommendations are generated.	8
35	EP-009-MS012	Integrate External Weather API to Enrich Diagnosis	As a Developer, I want to integrate an external weather API so that the diagnosis engine considers climate conditions when calculating water stress.	5
36	EP-009-US004	Monitor Remote Parcels Between Visits	As an agronomist, I want to monitor remote parcels between visits so that I can dispatch proactive recommendations upon critical environmental changes.	3
37	EP-009-US002	Adjust Client Irrigation Thresholds Remotely	As an agronomist, I want to adjust client irrigation thresholds remotely so that I can update crop recommendations without a physical field visit.	5
38	EP-009-US003	Generate Technical Report for Client	As an agronomist, I want to generate a technical report so that I can share monthly soil health summaries and recommendations in a professional format.	5
39	EP-009-US005	Configure Crop-Specific Thresholds as Agronomist Template	As an agronomist, I want to configure crop-specific thresholds as templates so that I can quickly apply standardized settings to new client parcels.	3
40	EP-008-US020	Manage Advanced Notification Preferences	As an agriculturist, I want to manage advanced notification preferences so that I	3

# Orden	User Story Id	Título	Descripción	Story Points (1 / 2 / 3 / 5 / 8)
			can route alerts only to the appropriate channels.	
41	EP-008-US021	Receive WhatsApp Alert with Direct Action Link	As an agriculturist, I want to receive WhatsApp alerts with direct action links so that I can address critical conditions immediately from my messaging app.	5
42	EP-008-TS018	Cloud RESTful API — Notifications Dispatch	As a Developer, I want to implement a multi-channel notification dispatch service in the backend so that internal modules can queue alerts for external delivery.	8
43	EP-006-US014	View and Manage Subscription Plan	As an agriculturist, I want to view and manage my subscription plan so that I can evaluate the current tier, benefits, and upgrade options.	3
44	EP-006-US015	Manage Billing and Payment Methods	As an agriculturist, I want to manage billing and payment methods so that I can access my payment history and update my credit cards.	3
45	EP-006-US016	Cancel Subscription and Request Hardware Return	As an agriculturist, I want to cancel my subscription and request a hardware return so that I can handle the offboarding process securely.	3
46	EP-006-TS017	Cloud RESTful API — Subscriptions & Billing	As a Developer, I want to implement subscription endpoints integrated with an external payment provider in the backend so that plans are managed automatically.	8
47	EP-010-US001	Monitor IoT Device Health Fleet	As a support staff member, I want to monitor the IoT device health fleet so that I can manage and maintain deployed devices proactively without unnecessary field visits.	5
48	EP-010-TS029	Cloud RESTful API — Fleet Health Endpoints	As a Developer, I want to implement fleet health endpoints in the backend so that the support SPA can retrieve and aggregate connection and battery statuses securely.	5

# Orden	User Story Id	Título	Descripción	Story Points (1 / 2 / 3 / 5 / 8)
49	EP-011-US001	Manage Customer Accounts from Admin Panel	As an operations specialist, I want to manage customer accounts from the admin panel so that the customer lifecycle and account suspensions can be controlled.	5
50	EP-011-US002	Manage and Activate IoT Devices from Admin Panel	As an operations specialist, I want to manage and activate IoT devices from the admin panel so that new hardware batches are registered and compromised devices are blocked.	5
51	EP-011-TS030	Cloud RESTful API — Admin Endpoints	As a Developer, I want to implement administrative endpoints in the backend so that the back-office SPA can securely manage accounts and device registries.	5
52	EP-012-US022	View Business Strategic Analytics	As a product owner, I want to review growth, retention, and churn metrics by segment so that the quarterly roadmap and strategies can be adjusted.	5
53	EP-012-US023	View Feature Adoption and Usage Heatmap	As a product owner, I want to see which features are used and how frequently by segment so that roadmap improvements can be prioritized based on real usage.	5
54	EP-012-TS031	Cloud RESTful API — Business Intelligence Endpoints	As a Developer, I want to implement analytical endpoints in the backend so that the BI SPA can retrieve aggregated usage and financial metrics.	8
55	EP-007-TS021	Implement Lazy Loading and Code Splitting for Dashboard Modules	As a Developer, I want the system to implement lazy loading of dashboard modules so that the initial load time is reduced.	5
56	EP-007-TS022	Implement Global State Management for Dashboard	As a Developer, I want to implement a global state manager on the web frontend so that sensor data, alerts, and configuration are synchronized between components without redundant API calls.	5

# Orden	User Story Id	Título	Descripción	Story Points (1 / 2 / 3 / 5 / 8)
57	EP-007- TS027	Internationalization Support	As a Developer, I want the system to support internationalization so that the platform can be used comfortably without language barriers.	3
58	EP-007- TS028	Accessibility Compliance	As a Developer, I want the system to comply with accessibility standards so that users with disabilities can access all features without barriers.	3
59	EP-005- MS001	Implement Async Event Processing Queue for IoT Telemetry	As a Developer, I want the system to implement an asynchronous message queue so that ingestion is decoupled from rule processing and data loss during load spikes is avoided.	8
60	EP-005- MS006	Implement Over-The-Air Firmware Update (OTA)	As a Developer, I want the system to support remote firmware updates so that improvements and security fixes are distributed without requiring field visits.	8
61	EP-007- MS002	Implement Auto-Scaling for Backend Services	As a Developer, I want to configure horizontal auto-scaling for backend services so that performance is maintained during peak loads without unnecessary costs during low-demand periods.	8
62	EP-007- MS003	Implement Data Retention and Archiving Policy	As a Developer, I want to implement a historical data retention and archiving policy so that storage costs are balanced with data availability for analysis.	5
63	EP-007- MS004	Implement Automated Backup and Recovery	As a Developer, I want to configure automatic backups of the main database so that data can be recovered after a catastrophic failure with minimal data loss.	8
64	EP-007- MS014	Setup CI/CD Pipeline for Modular Backend	As a Developer, I want to configure a CI/CD pipeline that compiles, tests, and deploys backend modules so that release time and deployment risk are reduced.	8

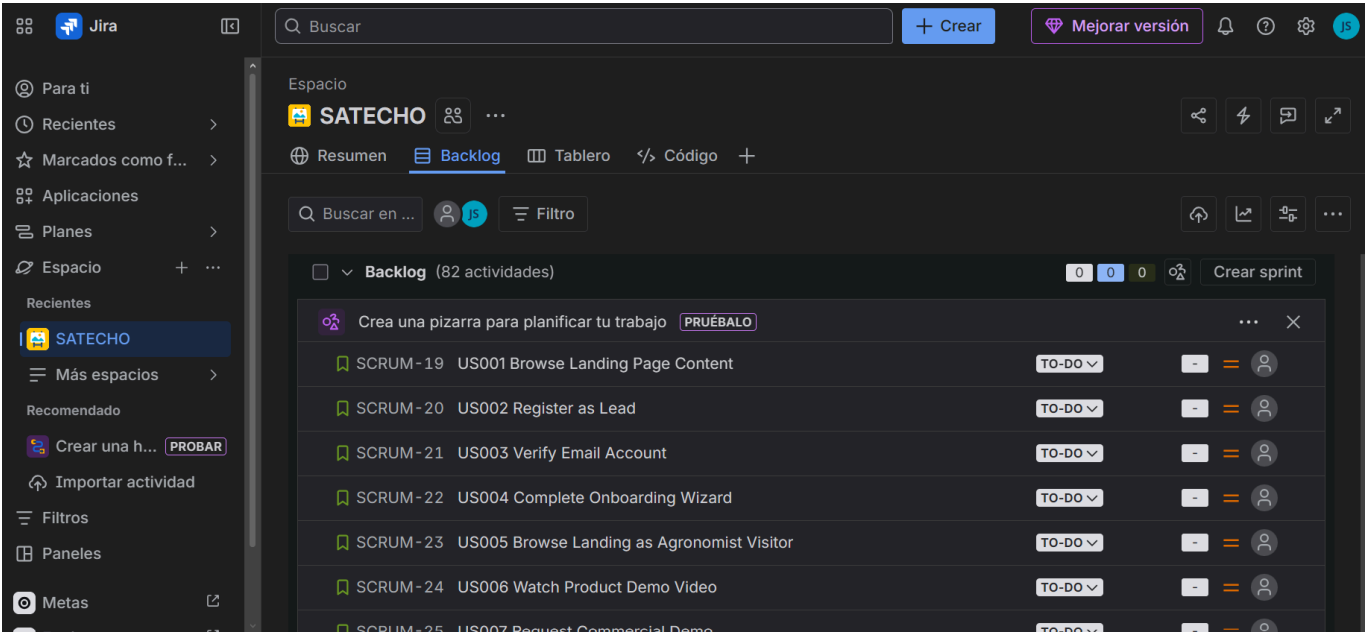
# Orden	User Story Id	Título	Descripción	Story Points (1 / 2 / 3 / 5 / 8)
65	EP-005-MS008	Implement Edge-to-Cloud Sync Protocol	As a Developer, I want to implement the Edge-to-Cloud synchronization protocol with delivery guarantee so that no data is lost during intermittent connectivity.	8
66	EP-005-MS009	Implement Edge API Authentication for IoT Devices	As a Developer, I want to implement mutual TLS authentication between ESP32 devices and the Edge component so that only authorized devices can communicate.	8
67	EP-005-MS010	Implement Local Rule Engine on Edge	As a Developer, I want the Edge to evaluate threshold rules locally so that critical actions are guaranteed even without cloud connectivity.	8
68	EP-005-MS011	Implement Edge Heartbeat and Health Reporting	As a Developer, I want the Edge to transmit periodic heartbeats so that offline nodes are detected automatically.	5
69	EP-005-MS013	Implement IoT Gateway Auto-Reconnection Protocol	As a Developer, I want to implement an automatic reconnection protocol with a command queue so that solenoid valve activation is guaranteed even with network intermittency.	8
70	EP-010-MS007	Implement Remote Hardware Diagnostics	As a Developer, I want to implement remote hardware diagnostics so that the support system can execute tests on the embedded sensors and valves directly from the cloud.	8
71	EP-002-TS026	Implement Biometric Auth for Critical Actions	As a Developer, I want the mobile app to require biometric authentication for critical actions so that accidental valve activations are prevented.	5
72	EP-004-MS005	Implement Zero-Touch Device Provisioning	As a Developer, I want the backend to support zero-touch provisioning so that hardware activation via QR scanning is secure and automated.	8

3.3. Product Backlog

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36	EP-009-US004	Monitor Remote Parcels Between Visits	As an agronomist, I want to monitor remote parcels between visits so that I can dispatch proactive recommendations upon critical environmental changes.	3
37	EP-009-US002	Adjust Client Irrigation Thresholds Remotely	As an agronomist, I want to adjust client irrigation thresholds remotely so that I can update crop recommendations without a physical field visit.	5
38	EP-009-US003	Generate Technical Report for Client	As an agronomist, I want to generate a technical report so that I can share monthly soil health summaries and recommendations in a professional format.	5
39	EP-009-US005	Configure Crop-Specific Thresholds as Agronomist Template	As an agronomist, I want to configure crop-specific thresholds as templates so that I can quickly apply standardized settings to new client parcels.	3
40	EP-008-US020	Manage Advanced Notification Preferences	As an agriculturist, I want to manage advanced notification preferences so that I	3

# Orden	User Story Id	Título	Descripción	Story Points (1 / 2 / 3 / 5 / 8)
			can route alerts only to the appropriate channels.	
41	EP-008-US021	Receive WhatsApp Alert with Direct Action Link	As an agriculturist, I want to receive WhatsApp alerts with direct action links so that I can address critical conditions immediately from my messaging app.	5
42	EP-008-TS018	Cloud RESTful API — Notifications Dispatch	As a Developer, I want to implement a multi-channel notification dispatch service in the backend so that internal modules can queue alerts for external delivery.	8
43	EP-006-US014	View and Manage Subscription Plan	As an agriculturist, I want to view and manage my subscription plan so that I can evaluate the current tier, benefits, and upgrade options.	3
44	EP-006-US015	Manage Billing and Payment Methods	As an agriculturist, I want to manage billing and payment methods so that I can access my payment history and update my credit cards.	3
45	EP-006-US016	Cancel Subscription and Request Hardware Return	As an agriculturist, I want to cancel my subscription and request a hardware return so that I can handle the offboarding process securely.	3
46	EP-006-TS017	Cloud RESTful API — Subscriptions & Billing	As a Developer, I want to implement subscription endpoints integrated with an external payment provider in the backend so that plans are managed automatically.	8
47	EP-010-US001	Monitor IoT Device Health Fleet	As a support staff member, I want to monitor the IoT device health fleet so that I can manage and maintain deployed devices proactively without unnecessary field visits.	5
48	EP-010-TS029	Cloud RESTful API — Fleet Health Endpoints	As a Developer, I want to implement fleet health endpoints in the backend so that the support SPA can retrieve and aggregate connection and battery statuses securely.	5

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49	EP-011-US001	Manage Customer Accounts from Admin Panel	As an operations specialist, I want to manage customer accounts from the admin panel so that the customer lifecycle and account suspensions can be controlled.	5
50	EP-011-US002	Manage and Activate IoT Devices from Admin Panel	As an operations specialist, I want to manage and activate IoT devices from the admin panel so that new hardware batches are registered and compromised devices are blocked.	5
51	EP-011-TS030	Cloud RESTful API — Admin Endpoints	As a Developer, I want to implement administrative endpoints in the backend so that the back-office SPA can securely manage accounts and device registries.	5
52	EP-012-US022	View Business Strategic Analytics	As a product owner, I want to review growth, retention, and churn metrics by segment so that the quarterly roadmap and strategies can be adjusted.	5
53	EP-012-US023	View Feature Adoption and Usage Heatmap	As a product owner, I want to see which features are used and how frequently by segment so that roadmap improvements can be prioritized based on real usage.	5
54	EP-012-TS031	Cloud RESTful API — Business Intelligence Endpoints	As a Developer, I want to implement analytical endpoints in the backend so that the BI SPA can retrieve aggregated usage and financial metrics.	8
55	EP-007-TS021	Implement Lazy Loading and Code Splitting for Dashboard Modules	As a Developer, I want the system to implement lazy loading of dashboard modules so that the initial load time is reduced.	5
56	EP-007-TS022	Implement Global State Management for Dashboard	As a Developer, I want to implement a global state manager on the web frontend so that sensor data, alerts, and configuration are synchronized between components without redundant API calls.	5

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57	EP-007- TS027	Internationalization Support	As a Developer, I want the system to support internationalization so that the platform can be used comfortably without language barriers.	3
58	EP-007- TS028	Accessibility Compliance	As a Developer, I want the system to comply with accessibility standards so that users with disabilities can access all features without barriers.	3
59	EP-005- MS001	Implement Async Event Processing Queue for IoT Telemetry	As a Developer, I want the system to implement an asynchronous message queue so that ingestion is decoupled from rule processing and data loss during load spikes is avoided.	8
60	EP-005- MS006	Implement Over-The-Air Firmware Update (OTA)	As a Developer, I want the system to support remote firmware updates so that improvements and security fixes are distributed without requiring field visits.	8
61	EP-007- MS002	Implement Auto-Scaling for Backend Services	As a Developer, I want to configure horizontal auto-scaling for backend services so that performance is maintained during peak loads without unnecessary costs during low-demand periods.	8
62	EP-007- MS003	Implement Data Retention and Archiving Policy	As a Developer, I want to implement a historical data retention and archiving policy so that storage costs are balanced with data availability for analysis.	5
63	EP-007- MS004	Implement Automated Backup and Recovery	As a Developer, I want to configure automatic backups of the main database so that data can be recovered after a catastrophic failure with minimal data loss.	8
64	EP-007- MS014	Setup CI/CD Pipeline for Modular Backend	As a Developer, I want to configure a CI/CD pipeline that compiles, tests, and deploys backend modules so that release time and deployment risk are reduced.	8

# Orden	User Story Id	Título	Descripción	Story Points (1 / 2 / 3 / 5 / 8)
65	EP-005-MS008	Implement Edge-to-Cloud Sync Protocol	As a Developer, I want to implement the Edge-to-Cloud synchronization protocol with delivery guarantee so that no data is lost during intermittent connectivity.	8
66	EP-005-MS009	Implement Edge API Authentication for IoT Devices	As a Developer, I want to implement mutual TLS authentication between ESP32 devices and the Edge component so that only authorized devices can communicate.	8
67	EP-005-MS010	Implement Local Rule Engine on Edge	As a Developer, I want the Edge to evaluate threshold rules locally so that critical actions are guaranteed even without cloud connectivity.	8
68	EP-005-MS011	Implement Edge Heartbeat and Health Reporting	As a Developer, I want the Edge to transmit periodic heartbeats so that offline nodes are detected automatically.	5
69	EP-005-MS013	Implement IoT Gateway Auto-Reconnection Protocol	As a Developer, I want to implement an automatic reconnection protocol with a command queue so that solenoid valve activation is guaranteed even with network intermittency.	8
70	EP-010-MS007	Implement Remote Hardware Diagnostics	As a Developer, I want to implement remote hardware diagnostics so that the support system can execute tests on the embedded sensors and valves directly from the cloud.	8
71	EP-002-TS026	Implement Biometric Auth for Critical Actions	As a Developer, I want the mobile app to require biometric authentication for critical actions so that accidental valve activations are prevented.	5
72	EP-004-MS005	Implement Zero-Touch Device Provisioning	As a Developer, I want the backend to support zero-touch provisioning so that hardware activation via QR scanning is secure and automated.	8